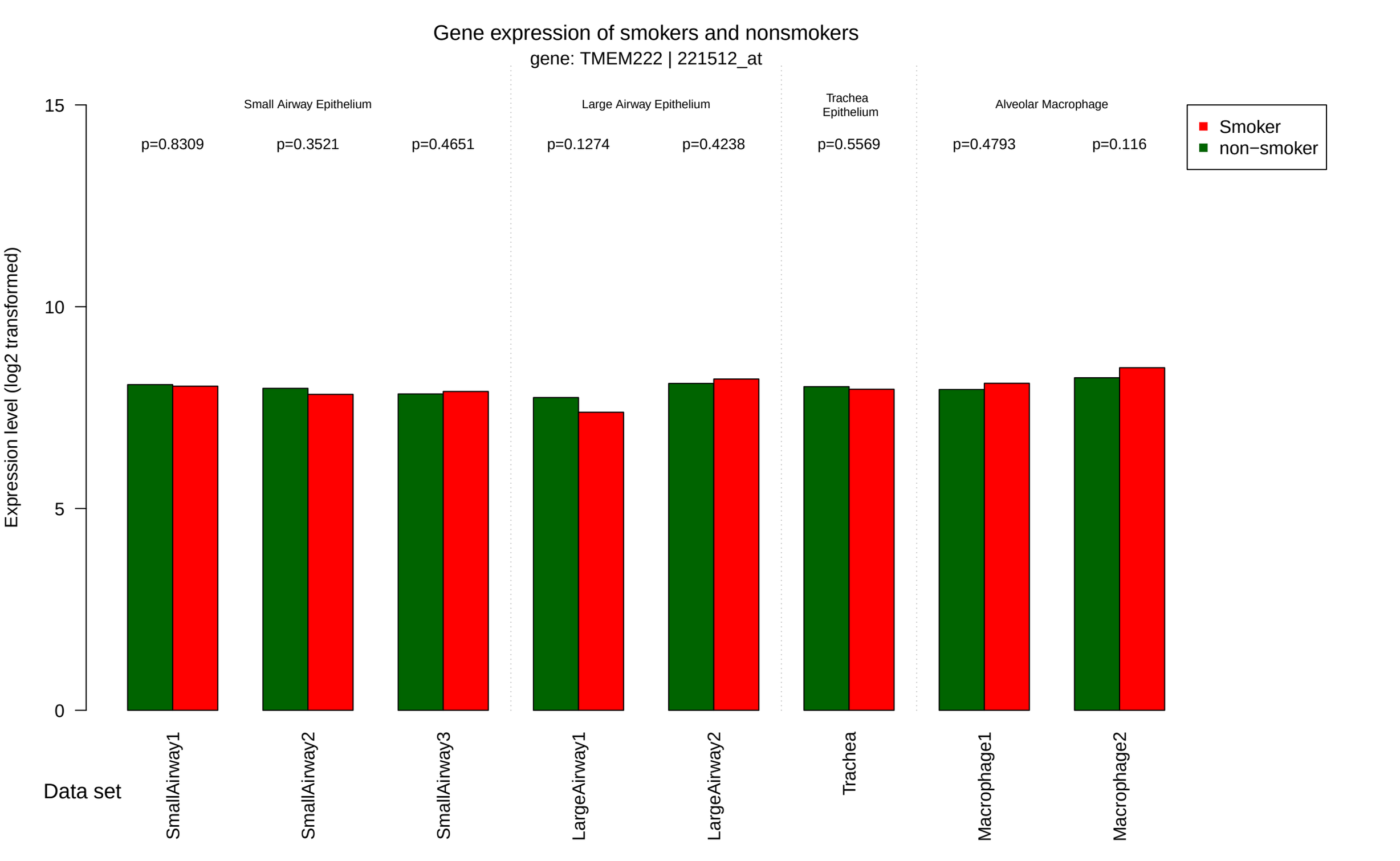
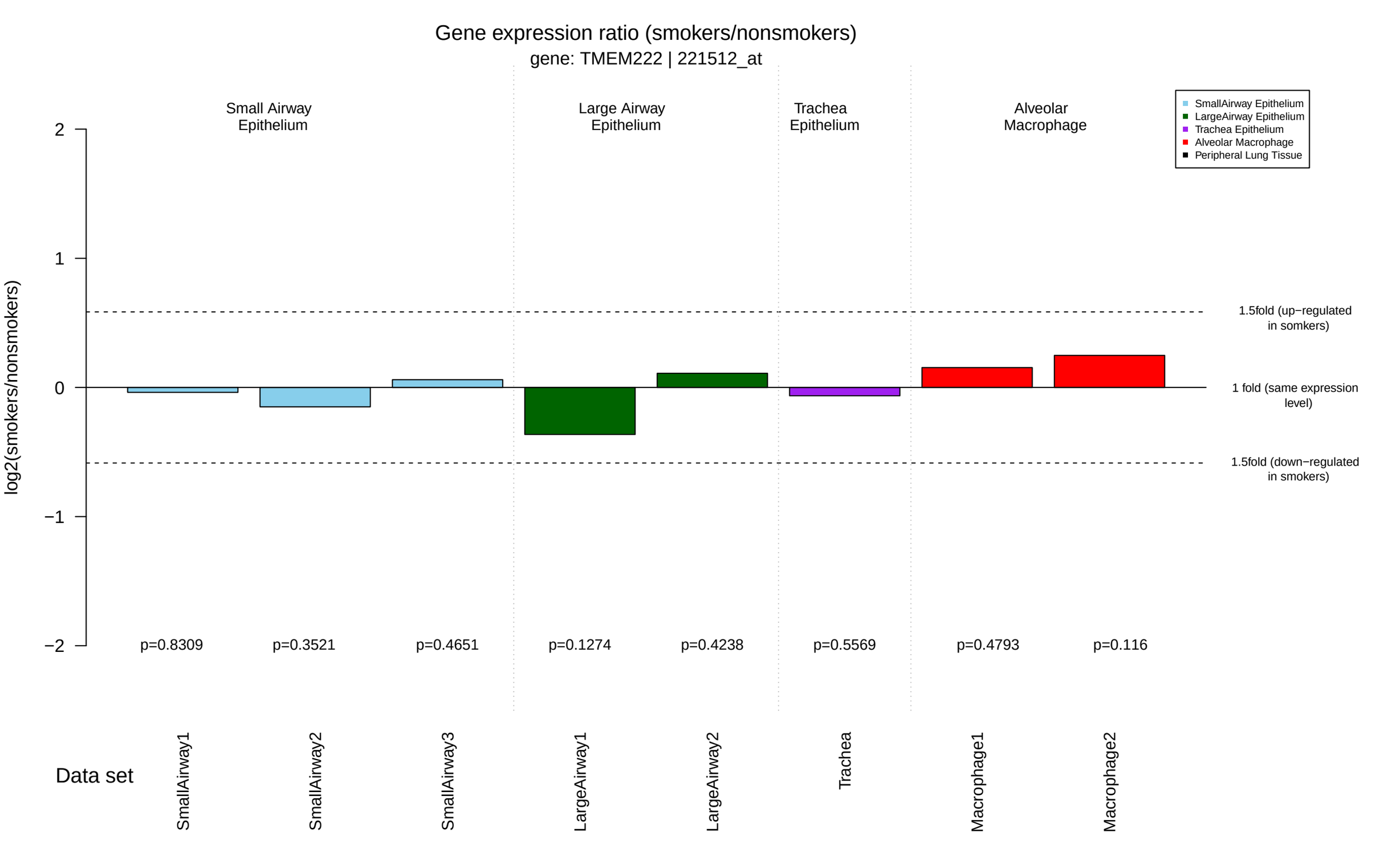
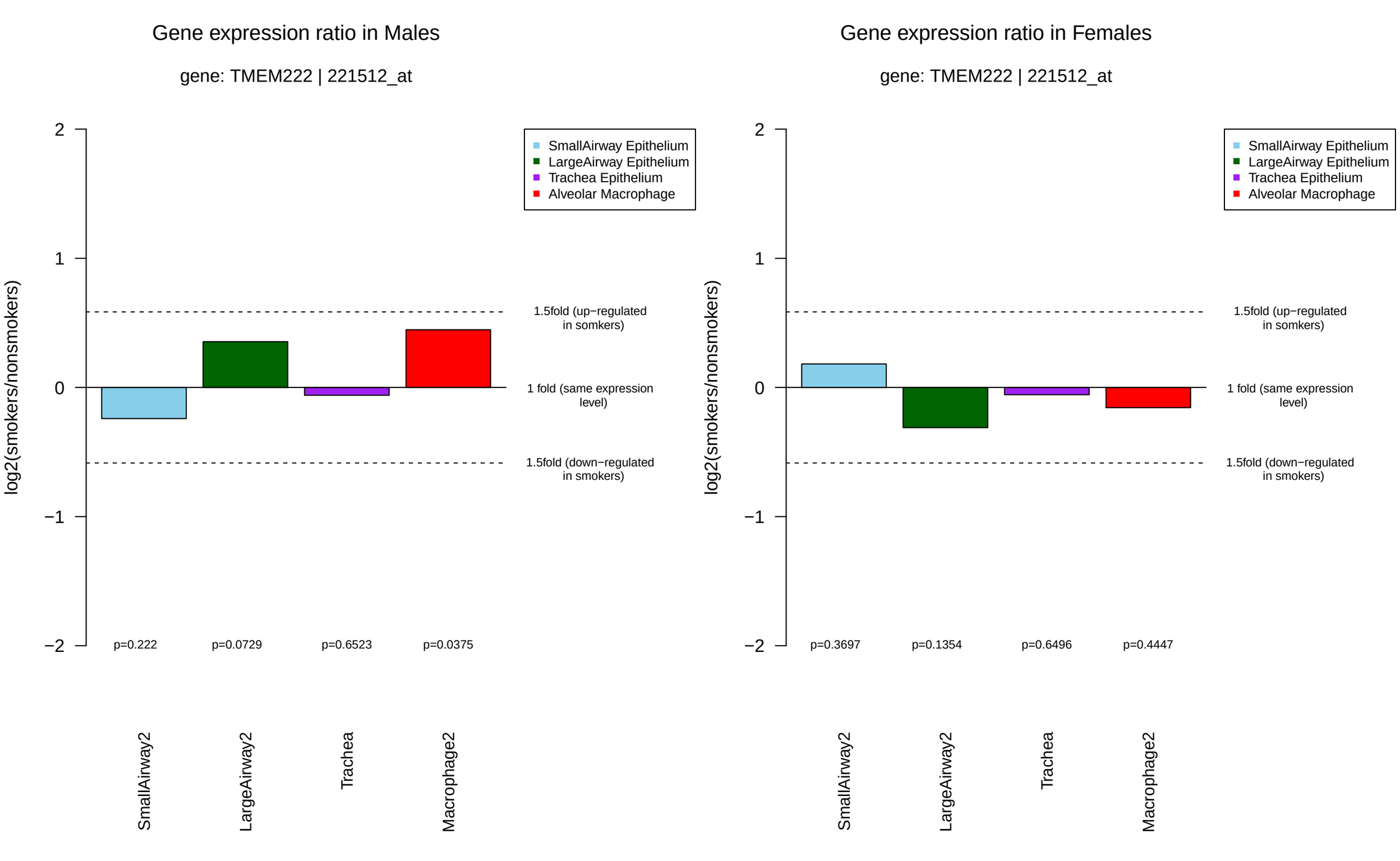
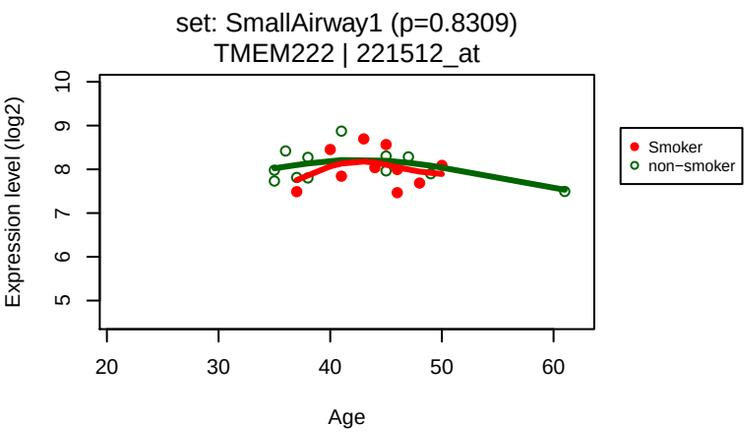
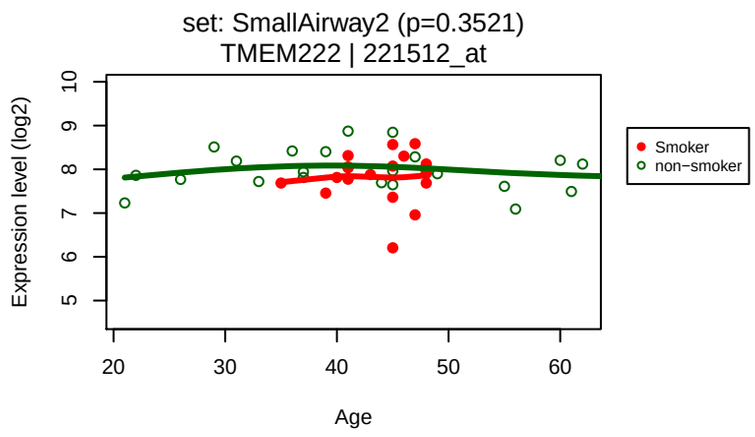
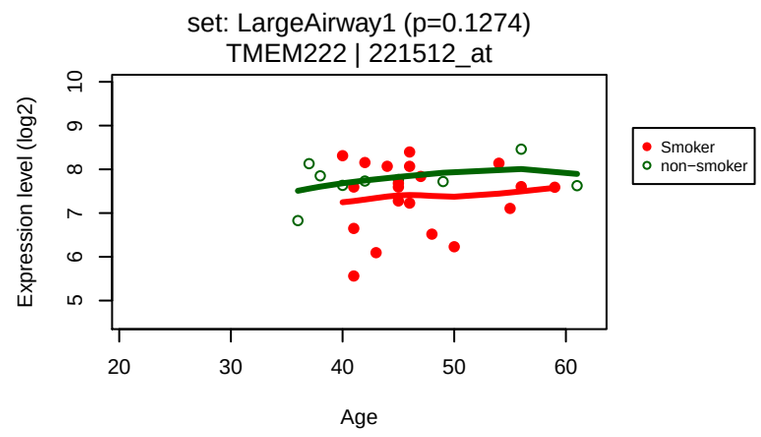

Gene expression of smokers and nonsmokers
gene: TMEM222 | 221512_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8309
p=0.3521
p=0.4651
p=0.1274
p=0.4238
p=0.5569
p=0.4793
p=0.116
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

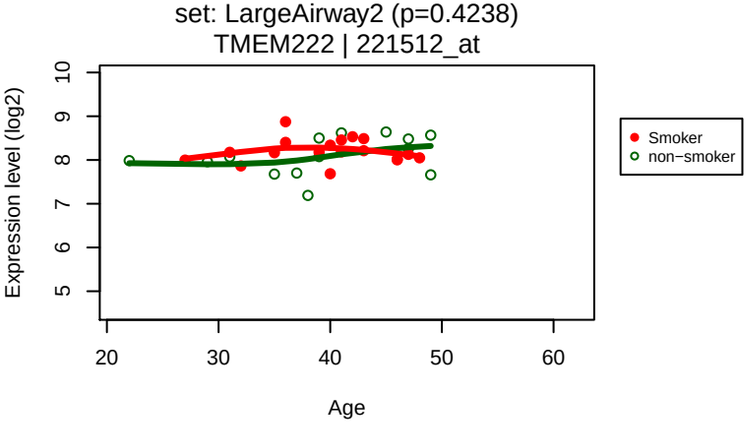

Gene expression ratio (smokers/nonsmokers)
gene: TMEM222 | 221512_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8309
p=0.3521
p=0.4651
p=0.1274
p=0.4238
p=0.5569
p=0.4793
p=0.116
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

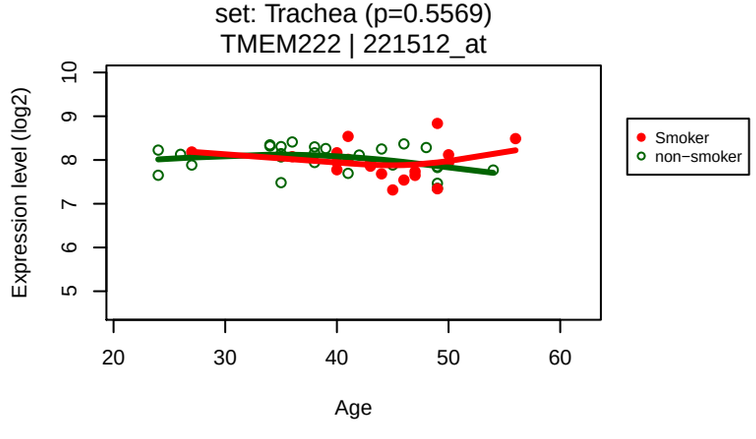

Gene expression ratio in Males
gene: TMEM222 | 221512_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.222
p=0.0729
p=0.6523
p=0.0375
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM222 | 221512_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3697
p=0.1354
p=0.6496
p=0.4447
SmallAirway2
LargeAirway2
Trachea
Macrophage2

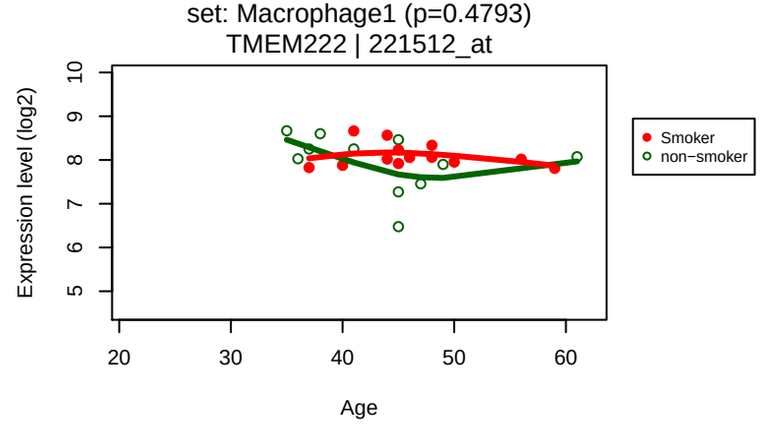

set: SmallAirway1 (p=0.8309)
TMEM222 | 221512_at
Expression level (log2)
Age
Smoker
non−smoker

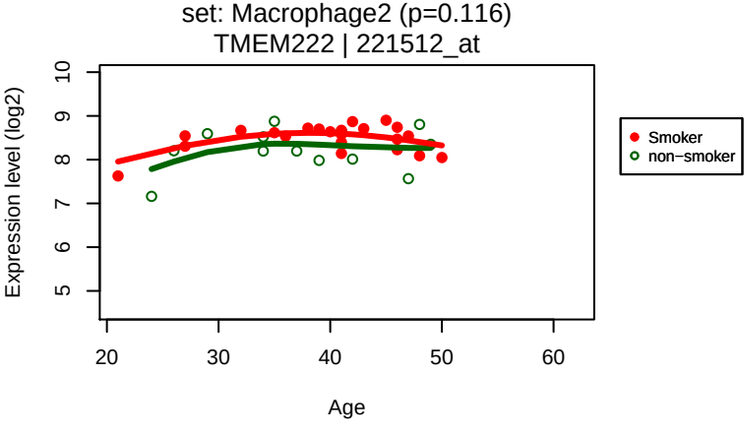

set: SmallAirway2 (p=0.3521)
TMEM222 | 221512_at
Expression level (log2)
Age
Smoker
non−smoker

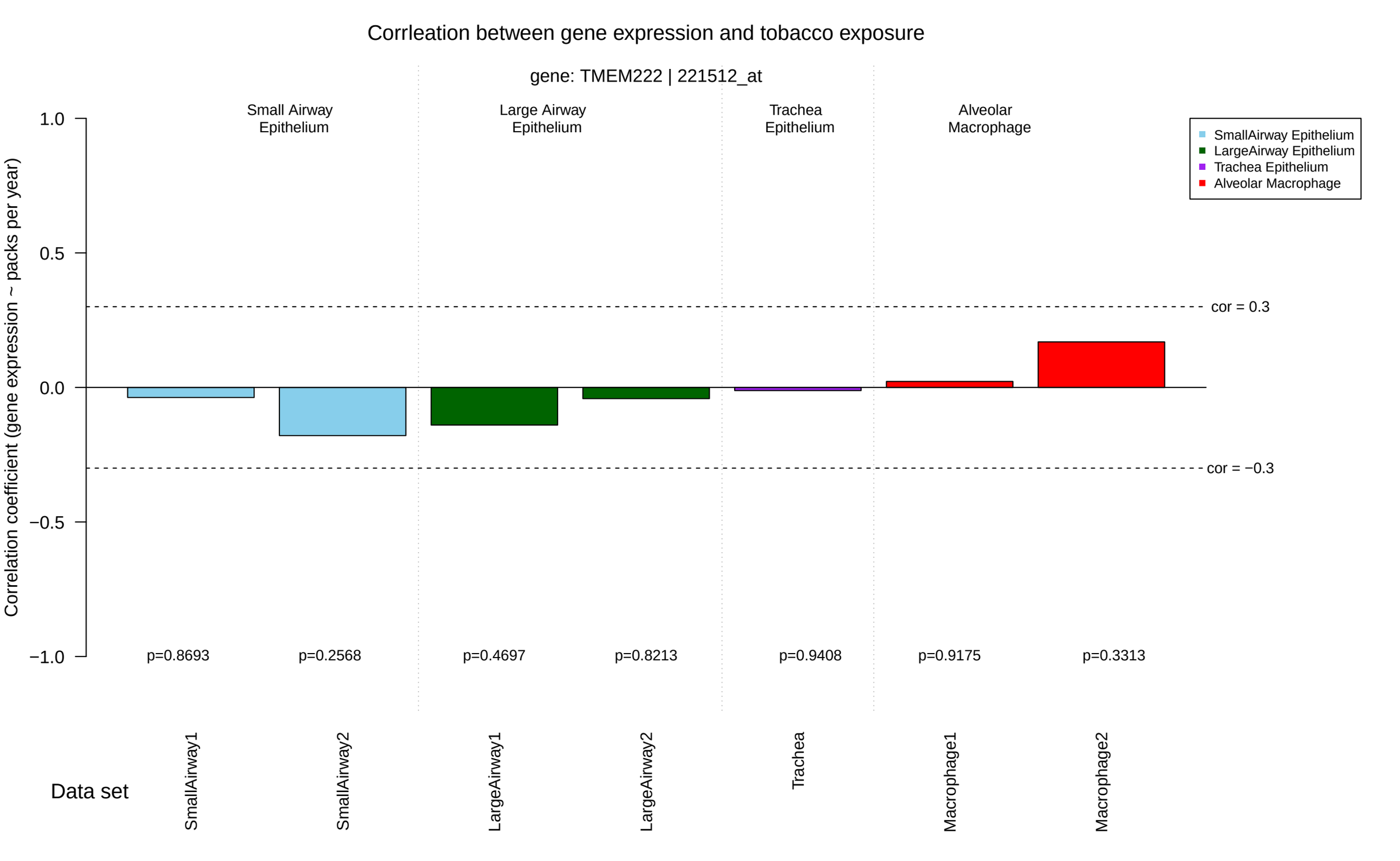

set: LargeAirway1 (p=0.1274)
TMEM222 | 221512_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4238)
TMEM222 | 221512_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5569)
TMEM222 | 221512_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4793)
TMEM222 | 221512_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.116)
TMEM222 | 221512_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM222 | 221512_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8693
p=0.2568
p=0.4697
p=0.8213
p=0.9408
p=0.9175
p=0.3313
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2